Assignment Report

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CSC 3324 02

HJ-OGH

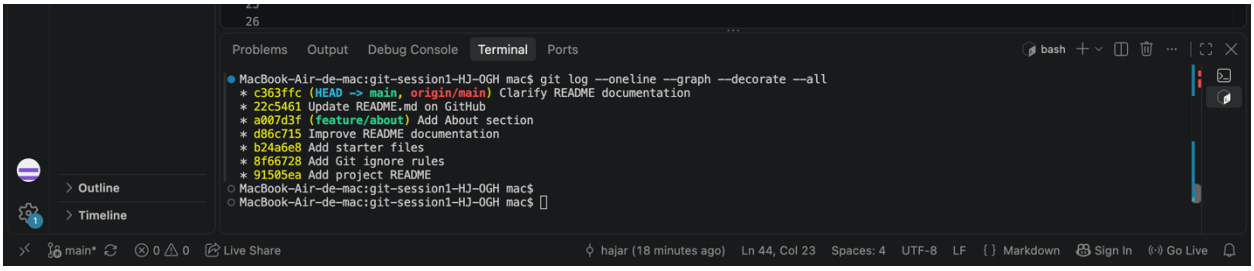
1. Repository setup

I used the provided starter files to set up my Git repository. I created and configured the .gitignore file to exclude local.env and ordinary .log files from version control while allowing important.log to remain tracked. I checked the repository status to verify which files were tracked, ignored, or modified.

2. README & Commit History

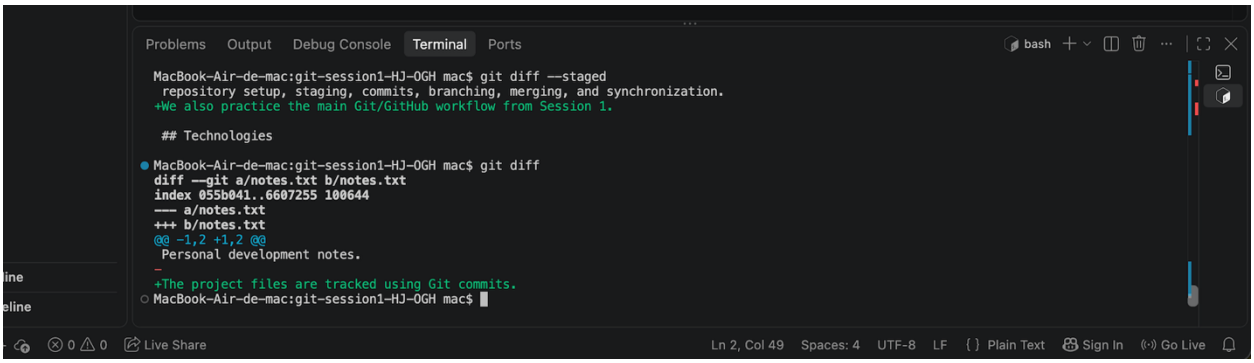
I created and updated README.md to describe the project. It includes my name, GitHub username, a short project description, the technologies used, a code example, a relevant link, and the provided project banner image.

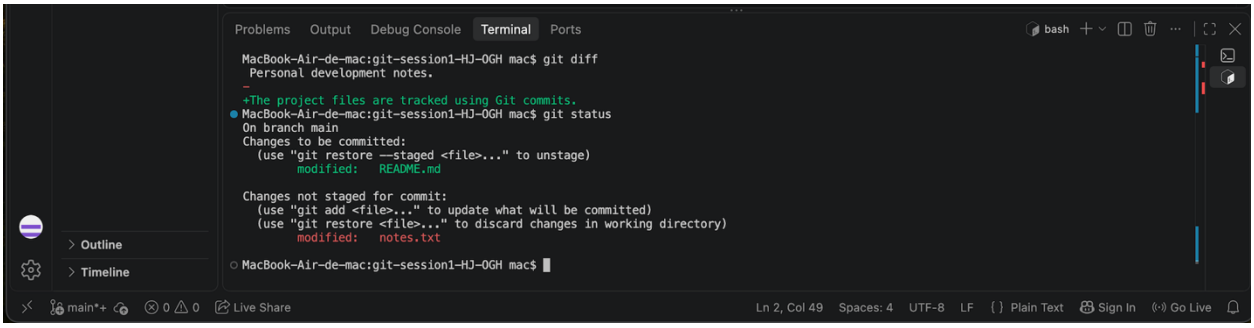
I made multiple meaningful commits to record the changes to my project. I used Git's commit history to review the changes and verify that the commits were recorded.



3. Selective Staging

To practice selective staging, I modified two tracked files: README.md and notes.txt. I staged only README.md, leaving the changes to notes.txt unstaged. I used git diff --staged to inspect the staged changes and git diff to inspect the unstaged changes.

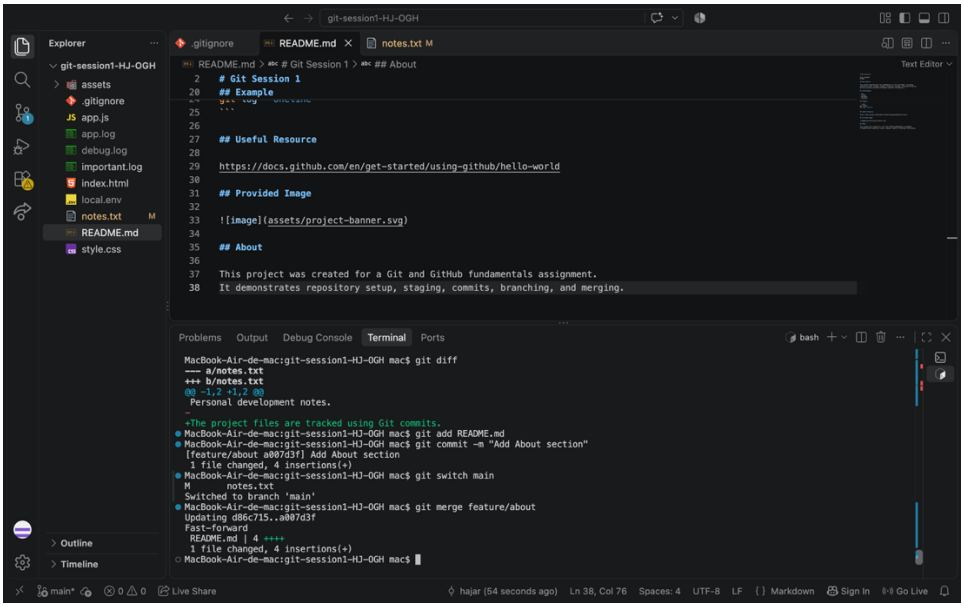




4. Branch and Merge

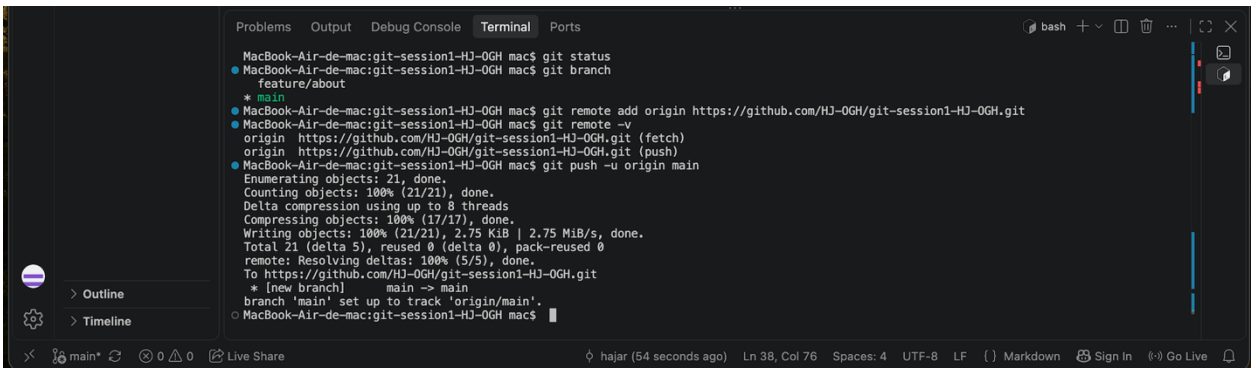
I created a feature branch named feature/about to add an About section to the README. I committed the changes on that branch, switched back to main, and merged the feature branch into main.

I reviewed the commit history to verify that the feature commit and merge were recorded.



5. Remote Changes: Push, Fetch, and Pull

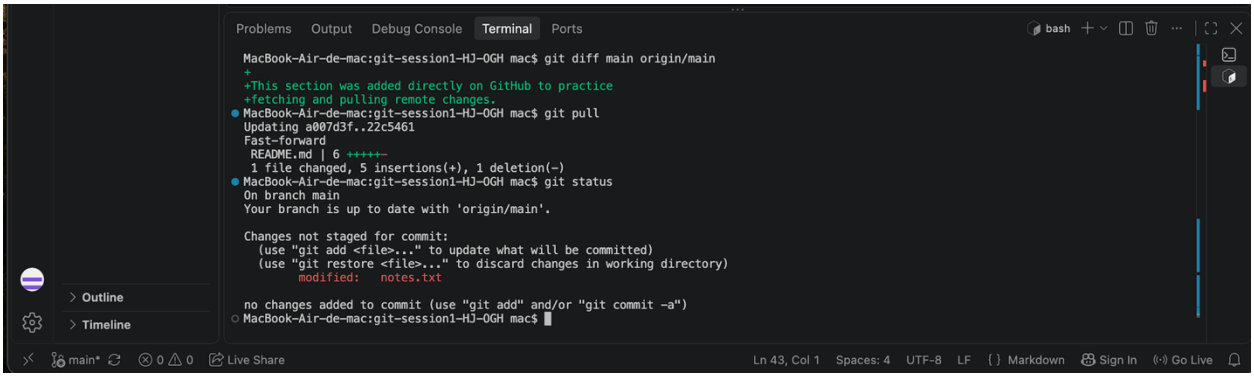
I created a GitHub repository and connected it to my local repository. I pushed the local main branch to GitHub so that the project files and commit history were available remotely.



Next, I edited README.md directly on GitHub and committed the remote change. I ran git fetch origin to retrieve information about the remote changes without immediately modifying my local working files. I inspected the difference between my local main branch and origin/main to verify that the remote branch contained a new commit.



Finally, I ran git pull to integrate the remote changes into my local branch and verified that the new README section was present locally.



6. Fair Use of AI

I used AI to understand some Git commands and to ask troubleshooting questions. I asked questions about commands such git diff, and git fetch, as well as how to check my commits and undo or stage changes correctly. I used the explanations that helped me understand what to do and adapted the commands to my own files and repository. I checked the results by running the commands in my terminal, looking at the commit history, and checking my repository on GitHub.

7. Sources and Reflection

Sources:

GitHub documentation: https://docs.github.com/

The hardest concept for me was understanding the difference between staging, committing, fetching, and pulling changes. If Git behaves unexpectedly, I would first check git status to see which branch I am on and whether I have staged, unstaged, or uncommitted changes, then check the relevant command output or commit history to understand what happened.